Wholeness as Lossless Causal Compression

A three-phase coding theorem for vanishing activity,
with uniform finite approximations and a last-event entropy law

Jan Mikulík
Independent mathematical note
https://github.com/jendamikulik/finite-Shannon-theorem

20 October 2026 *Mathematical draft; not a Lean-verified extension*

Abstract

We study a single random seed that realizes, exactly and with shared-history consistency, every deterministic adaptive policy on a locally finite binary controlled tree. At time t, every conditional success probability is bounded by an attainable envelope $p_t \in (0, 1/2]$. An explicit geometric-gated seed proves

$$\sum_{t=1}^{n} h_2(p_t) \le C_n \le \sum_{t=1}^{n} h_2(p_t) + 2\sum_{t=1}^{n} p_t,$$

uniformly in the number of policies and in the history dependence of the kernels. We distinguish three exact infinite-horizon regimes: no countable seed, a countable seed of necessarily infinite entropy, and a countable seed of finite entropy. The same trichotomy has an operational coding meaning: respectively, no almost-surely finite binary description, finite descriptions with necessarily infinite expected length, and finite expected length. We prove a zero–one law for transcript atomic capacity and construct finite strategy approximations whose total variation error is at most the tail sum $\sum_{t>N} p_t$, simultaneously for all policies. For $p_t = 1/[t(\ln t)^2]$, the finite-horizon optimal entropy is $C_n = \log_2 \ln n + O(1)$; moreover the entropy of the last successful time satisfies $H(J_n) = \log_2 \ln n + \kappa + o(1)$ for an explicit convergent series expression for κ. The model formalizes losslessness as exact preservation of policy laws, not invertibility of trajectories or lossless encoding of the transition kernels. Publication priority and formal verification of the new results remain open.

1 What "wholeness" means—and what it does not

A controlled process carries a family of distributions indexed by policies. A *whole causal representation* is one distribution over deterministic response tables whose projection through every policy reproduces that policy's entire output law, with the same answer whenever two policies share a history. This is a compatibility property of *all counterfactual readouts at once*, not merely a coupling of separately listed marginals.

The word *lossless* here has a precise, limited meaning: every finite transcript has its prescribed probability. It does *not* assert bijective recovery of independent simulations for different policies, finite Kolmogorov complexity, efficient computation of the decoder, or any physical claim about consciousness or the substrate of the universe. The kernels and policy rules are assumed known to the decoder and are not charged to the entropy budget. This distinction is essential in interpreting the quantitative theorems below.

2 Controlled trees and common exact seeds

Let time range over $t \in \mathbb{N}_{\ge 1}$. At each finite action–output history h of length $t-1$, a finite nonempty action set $\mathcal{A}_t(h)$ is available. After choosing $a \in \mathcal{A}_t(h)$, the environment returns a bit $Y_t \in \{0, 1\}$ with

$$\mathbb{P}(Y_t = 1 \mid h, a) = q_t(h, a).$$

We impose the following uniform, attainable envelopes:

$$0 \le q_t(h,a) \le p_t \le \tfrac{1}{2}, \qquad p_t > 0, \qquad \max_{a \in \mathcal{A}_t(h)} q_t(h,a) = p_t \quad \text{for every } t, h. \tag{1}$$

The condition on the maximum allows the maximizing action to depend on history; no stationarity is assumed. A deterministic adaptive policy π selects an available action from the history of observed outputs. A *total response strategy* f specifies a bit $f_t(h,a)$ for every finite history and every available action, including counterfactual nodes.

Definition 2.1 (Exact common causal seed)**.** *A random variable S and a deterministic map $S \mapsto f_S$ form an* exact common causal seed *when, for every deterministic policy π and every finite action–output transcript compatible with π, the probability of that transcript under f_S equals the product of its specified conditional transition probabilities. Equivalently, the law of the readout $D_\pi(S)$ equals the process law P_π for every π, with D_π induced by the* same *response table f_S.*

For $n \ge 1$, let C_n be the infimum of the Shannon entropy (in bits) of countably supported common seeds exact through time n. As the depth-n tree has finitely many nodes, the response table restricted to it has only finitely many values, so this is also the infimum over finite seeds. For the infinite horizon, we distinguish (a) existence of a countably supported exact seed and (b) existence of one with finite entropy. Write $h_2(u) = -u \log_2 u - (1-u)\log_2(1-u)$, with $0 \log 0 = 0$.

3 A universal sparse causal construction

Theorem 3.1 (Geometric-gated causal embedding)**.** *Under (1), there is a single exact causal seed S, defined on a product of countable alphabets, with independent coordinates Z_t satisfying*

$$\mathbb{P}(Z_t = 0) = 1 - p_t, \qquad \mathbb{P}(Z_t = k) = p_t 2^{-k} \quad (k \ge 1). \tag{2}$$

The coordinate entropy is

$$H(Z_t) = h_2(p_t) + 2p_t, \tag{3}$$

and for every horizon n,

$$\boxed{\sum_{t=1}^{n} h_2(p_t) \le C_n \le \sum_{t=1}^{n} h_2(p_t) + 2\sum_{t=1}^{n} p_t.} \tag{4}$$

If $\sum_t p_t < \infty$, then S is supported by a countable set of response strategies and, with probability one, there is a finite random $T(S)$ after which all *policies return only zeros.*

Proof. For each t, h, a, choose a binary expansion (e.g. the expansion terminating in zeros at dyadic rationals)

$$\frac{q_t(h,a)}{p_t} = \sum_{k \ge 1} b_{t,k}(h,a) 2^{-k}, \qquad b_{t,k}(h,a) \in \{0,1\}.$$

We may use the all-ones expansion for 1. Define the same total response strategy for every policy by

$$f_{Z,t}(h,a) = \begin{cases} 0, & Z_t = 0, \\ b_{t,Z_t}(h,a), & Z_t \ge 1. \end{cases}$$

Its marginal response probability is exactly $p_t \sum_{k \ge 1} b_{t,k}(h,a) 2^{-k} = q_t(h,a)$. The random history before time t is measurable with respect to $Z_1, \dots, Z_{t-1}$, which is independent of Z_t. Thus these are also the *conditional* response probabilities after any attainable history; induction gives every prescribed transcript probability. The same function f_Z is consulted by every policy, establishing shared-history consistency.

Let G have mass $\mathbb{P}(G = k) = 2^{-k}$, $k \geq 1$. Then $H(G) = \sum_{k\geq 1} k2^{-k} = 2$. Conditioning on the indicator $\mathbf{1}_{Z_t \neq 0}$ gives (3) by the entropy chain rule. For the upper bound in (4), push the law of $(Z_1, \ldots, Z_n)$ forward to the finite set of depth-n response tables. Entropy cannot increase under a deterministic mapping, so its entropy is at most $\sum_{t\leq n} H(Z_t)$.

For the lower bound, at every visited history select an action attaining p_t. Under the corresponding deterministic maximizing policy $\pi^\star$, the output bits are independent $B_t \sim \text{Bernoulli}(p_t)$, by induction on the history. Their transcript is a deterministic readout of any common seed. Data processing therefore yields $H(S) \geq H(B_1, \ldots, B_n) = \sum_{t\leq n} h_2(p_t)$.

Finally, if $\sum p_t < \infty$, the first Borel–Cantelli lemma implies that $Z_t \neq 0$ at only finitely many times almost surely. The sequences with finitely many nonzero coordinates in $\{0, 1, 2, \ldots\}^{\mathbb{N}}$ form a countable set, giving a countable seed after a harmless re-enumeration. Whenever all late coordinates are zero, the definition of f_Z makes all late responses zero at *every* history and action. □

Remark 3.2 (Uniform sparsity)**.** *For any policy, its total number of ones is bounded, under the common coupling, by* $\sum_t \mathbf{1}_{\{Z_t \neq 0\}}$*. Hence* $\sup_\pi \mathbb{E} \sum_t Y_t \leq \sum_t p_t$*. The random last active* time*, however, need not have finite expectation.*

4 Three exact coding regimes

Theorem 4.1 (Infinite-horizon trichotomy)**.** *Assume* (1)*. Exactly one of the following alternatives holds:*

1. *If* $\sum_t p_t = \infty$*, no countably supported exact common causal seed exists.*
2. *If* $\sum_t p_t < \infty$ *but* $\sum_t h_2(p_t) = \infty$*, a countably supported exact seed exists, but* every *exact common seed with countable support has infinite Shannon entropy.*
3. *If* $\sum_t h_2(p_t) < \infty$*, an exact countably supported common seed of finite Shannon entropy exists.*

Moreover, the three alternatives are equivalent, respectively, to: (i) no almost-surely finite, exactly decoded binary description of the common seed; (ii) almost-surely finite descriptions exist, but every exact prefix description has infinite expected length; and (iii) an exact prefix description of finite expected length exists.

Proof. Under $\pi^\star$, every prescribed length-n output string y has probability

$$\prod_{t=1}^{n} p_t^{y_t}(1 - p_t)^{1-y_t} \leq \prod_{t=1}^{n}(1 - p_t).$$

If $\sum p_t = \infty$, the right side tends to zero. Thus the entire infinite product output law is non-atomic: any individual infinite sequence lies in cylinders of vanishing mass. The deterministic image of a countably supported seed is countably supported and must possess an atom, a contradiction.

When $\sum p_t < \infty$, Theorem 3.1 supplies a countably supported exact seed. For every exact seed S and every n, $H(S) \geq \sum_{t\leq n} h_2(p_t)$. If that sum diverges, $H(S) = \infty$. If it converges, then $h_2(p_t) \geq p_t$ on $(0, 1/2]$ ensures $\sum p_t < \infty$, and the explicit seed satisfies

$$H(S) \leq \sum_{t\geq 1}[h_2(p_t) + 2p_t] < \infty.$$

For this countable-valued tuple, the entropy is the supremum of the entropies of its finite-prefix partitions (the partitions refine towards its point atoms). Independence then identifies that supremum with $\sum_t H(Z_t)$, proving the claimed finite-entropy bound.

For the coding statements, the set of all finite binary words is countable, so a distribution exactly reconstructed from a finite word almost surely must be countably supported. Conversely, enumerate the atoms of any countable seed with weights $w_i > 0$. The integer lengths $\ell_i = \lceil -\log_2 w_i \rceil$ obey

$\sum_i 2^{-\ell_i} \le \sum_i w_i = 1$. The countable Kraft inequality therefore provides an injective prefix code with every ℓ_i finite, and when $H(S) < \infty$ its mean length obeys

$$H(S) \le \mathbb{E}\ell(S) < H(S) + 1.$$

For any prefix code, the Shannon lower bound $H(S) \le \mathbb{E}\ell(S)$ holds also in the extended-real sense; thus if $H(S) = \infty$, every exact prefix code has infinite mean length. □

Remark 4.2 (Which kind of finite?)**.** *Alternative (ii) is the precise separation between* finite for every realized codeword *and* finite on average*. It does not refer to how many different policies exist. In general the latter family is uncountable even though the seed has countable support.*

5 Atomic capacity and uniform finite approximation

For a discrete distribution P, let $A_K(P)$ be the mass of its K largest atoms (or all of its atoms if fewer than K exist). Let $P_{n,\pi}$ be the law of the length-n transcript under policy π. Define its *uniform atomic capacity* by

$$\mathfrak{a} = \sup_{K \ge 1} \inf_{n \ge 1,\, \pi} A_K(P_{n,\pi}). \tag{5}$$

Theorem 5.1 (Zero–one atomic-capacity law)**.** *Under* (1)*,*

$$\boxed{\mathfrak{a} = \begin{cases} 0, & \sum_t p_t = \infty, \\ 1, & \sum_t p_t < \infty. \end{cases}}$$

Consequently $\mathfrak{a} = 1$ is compatible with an exact seed of necessarily infinite Shannon entropy.

Proof. If $\sum p_t = \infty$, use $\pi^\star$. Every length-n transcript atom has mass at most $r_n = \prod_{t \le n}(1 - p_t)$, so $A_K(P_{n,\pi^\star}) \le K r_n \to 0$ for each fixed K, forcing $\mathfrak{a} = 0$.

If $\sum p_t < \infty$, let $\varepsilon > 0$ and choose N with $\sum_{t>N} p_t \le \varepsilon$. For any deterministic policy and any $n > N$, the event that all outputs from $N+1$ through n are zero has probability at least $1 - \varepsilon$, by a union bound. At most 2^N length-n transcripts can occur on this event (the policy determines its actions from the output history). The same bound is trivial when $n \le N$. Thus $A_{2^N}(P_{n,\pi}) \ge 1 - \varepsilon$ uniformly in n, π, so $\mathfrak{a} = 1$. □

The next result makes the meaning of compression uniform in the policy, rather than merely pointwise in π.

Theorem 5.2 (Finite approximants for the entire strategy law)**.** *Suppose $\sum_t p_t < \infty$. For every N there is a* finitely supported *distribution Q_N over infinite total response strategies, with entropy at most*

$$H(Q_N) \le \sum_{t \le N} \big(h_2(p_t) + 2p_t\big), \tag{6}$$

such that the exact common-strategy distribution Q of Theorem 3.1 obeys

$$\mathrm{TV}(Q, Q_N) \le \sum_{t > N} p_t. \tag{7}$$

In particular, letting P_π and $P_\pi^{(N)}$ denote the associated infinite transcript laws,

$$\sup_\pi \mathrm{TV}(P_\pi, P_\pi^{(N)}) \le \sum_{t > N} p_t.$$

There is a single coupling *under which, with probability at least $1 - \sum_{t>N} p_t$, all policies simultaneously produce identical infinite transcripts in the exact and the approximating models.*

Proof. Sample the coordinates Z_t as in (2). Let $f_Z^{(N)}$ agree with f_Z through time N and return zero at every node at later times. The law Q_N of $f_Z^{(N)}$ has finite support: at any fixed finite depth there are only finitely many history–action nodes and thus finitely many binary response tables, regardless of the countable alphabet of each Z_t. The table $f_Z^{(N)}$ depends only on $Z_1, \dots, Z_N$, which proves (6) by data processing. If $Z_t = 0$ for every $t > N$, then the entire infinite response tables f_Z and $f_Z^{(N)}$ coincide, simultaneously for every history, action and policy. A union bound gives

$$\mathbb{P}(f_Z \neq f_Z^{(N)}) \leq \mathbb{P}(\exists t > N : Z_t \neq 0) \leq \sum_{t>N} p_t.$$

The coupling bound for total variation and its contraction under policy readout prove the result. □

Remark 5.3 (Exactness versus approximation)**.** *Q_N is exactly causal and exactly matches the original kernels* through time *N; beyond N it deliberately returns zero, so the approximation to the infinite-horizon law is not exact. The infinite law Q* is *exact. The distinction is never suppressed by the phrase "lossless compression."*

6 The last-event entropy law

Assume the initial time is 3 and consider the maximizing policy:

$$B_t \overset{\text{ind}}{\sim} \text{Bernoulli}(p_t), \qquad J_n = \max\big(\{t \in \{3, \dots, n\} : B_t = 1\} \cup \{0\}\big).$$

Lemma 6.1 (An exact identity)**.** *For any positive $p_3, \dots, p_n < 1$, one has*

$$\boxed{\sum_{t=3}^{n} h_2(p_t) - H(J_n) = \sum_{s=3}^{n-1} h_2(p_s)\Big(1 - \prod_{t=s+1}^{n} (1 - p_t)\Big).} \tag{8}$$

Proof. J_n is a deterministic function of $B_3, \dots, B_n$, so the left-hand side equals $H(B_3, \dots, B_n \mid J_n)$. Conditional on $J_n = j \geq 3$, the outcomes at j and all later times are fixed, whereas the earlier independent Bernoulli variables preserve their original joint law. Consequently,

$$H(B_3, \dots, B_n \mid J_n) = \sum_{j=3}^{n} \mathbb{P}(J_n = j) \sum_{s=3}^{j-1} h_2(p_s).$$

Exchanging finite sums and observing that $\mathbb{P}(J_n > s) = 1 - \prod_{t=s+1}^{n}(1 - p_t)$ gives (8). □

Theorem 6.2 (Critical double-logarithmic regime)**.** *Let*

$$p_t = \frac{1}{t(\ln t)^2}, \qquad t \geq 3.$$

Then $\sum_{t\geq 3} p_t < \infty$ while $\sum_{t\geq 3} h_2(p_t) = \infty$. Nevertheless every policy has only finitely many successes almost surely under one shared seed, and $\sup_\pi \mathbb{E} \sum_{t\geq 3} Y_t < \infty$. The optimal finite-horizon cost satisfies

$$\boxed{C_n = \log_2 \ln n + O(1).} \tag{9}$$

More precisely, for some finite real number κ,

$$\boxed{H(J_n) = \log_2 \ln n + \kappa + o(1).} \tag{10}$$

Let

$$\kappa_0 = \lim_{n\to\infty} \Big(\sum_{t=3}^{n} h_2(p_t) - \log_2 \ln n\Big),$$

$$D_\infty = \sum_{s=3}^{\infty} h_2(p_s)\Big(1 - \prod_{t=s+1}^{\infty} (1 - p_t)\Big).$$

Both are finite, and $\kappa = \kappa_0 - D_\infty$. For the maximizing policy, the last-success time J_∞ is finite almost surely, but $\mathbb{E}J_\infty = \infty$ and $H(J_\infty) = \infty$.

Proof. The integral test gives $\sum_{t\geq 3} p_t < \infty$ and the tail relation

$$\sum_{t\geq m} p_t = \frac{1}{\ln m} + O\Big(\frac{1}{m(\ln m)^2}\Big). \tag{11}$$

As $u \downarrow 0$, $h_2(u) = u\log_2(1/u) + O(u)$. Therefore

$$h_2(p_t) = \frac{1}{\ln 2}\frac{1}{t\ln t} + O\left(\frac{1+\ln\ln t}{t(\ln t)^2}\right). \tag{12}$$

The error series converges; by the integral test, $\sum_{t=3}^{n}(t\ln t)^{-1} - \ln\ln n$ converges as well. Hence κ_0 exists, and the first entropy sum equals $\log_2 \ln n + \kappa_0 + o(1)$. Applying Theorem 3.1, the two sides of (4) differ by at most $2\sum_{t\geq 3} p_t < \infty$. This proves (9), simultaneous quiescence, and the expected-count assertion.

Let D_n denote the right-hand side of (8). It increases with n. The union bound and (11) give

$$D_n \leq \sum_{s=3}^{\infty} h_2(p_s)\sum_{t>s} p_t \leq O(1) + O\left(\sum_{s\geq s_0}\frac{1}{s(\ln s)^2}\right) < \infty.$$

Thus monotone convergence yields $D_n \uparrow D_\infty < \infty$. Subtracting the identity (8) proves (10) and the stated formula for κ.

Finally, the probability of some success at or after time m is

$$\mathbb{P}(J_\infty \geq m) = 1 - \prod_{t\geq m}(1-p_t) \sim \sum_{t\geq m} p_t \sim \frac{1}{\ln m}.$$

This follows because $\sum_{t\geq m} p_t \to 0$ and $1 - \prod(1-p_t)$ is asymptotic to this tail sum. Thus $J_\infty < \infty$ almost surely but $\mathbb{E}J_\infty = \sum_{m\geq 1}\mathbb{P}(J_\infty \geq m) = \infty$. For $j \geq 3$, $\mathbb{P}(J_\infty = j) = p_j\prod_{t>j}(1-p_t) \sim p_j$; therefore the Shannon terms are asymptotic to $p_j\log_2(1/p_j)$, whose sum diverges by (12). This proves $H(J_\infty) = \infty$. □

Corollary 6.3 (A uniform finite-seed accuracy–entropy tradeoff)**.** *In the critical model, for all sufficiently small $\varepsilon > 0$ there is a* finite *strategy distribution Q_ε such that*

$$\sup_{\pi} \mathrm{TV}(P_\pi, P_{\pi,\varepsilon}) \leq \varepsilon, \qquad H(Q_\varepsilon) \leq \log_2(1/\varepsilon) + O(1).$$

Proof. Choose $N = \lceil e^{1/\varepsilon}\rceil$. By monotonicity of $x \mapsto 1/[x(\ln x)^2]$, the tail bound $\sum_{t>N} p_t \leq \int_N^\infty dx/[x(\ln x)^2] = 1/\ln N \leq \varepsilon$ holds. Theorem 5.2 yields a finite common seed with entropy at most $\sum_{t=3}^{N}(h_2(p_t) + 2p_t) = \log_2\ln N + O(1) = \log_2(1/\varepsilon) + O(1)$. □

7 A phase diagram of time-inhomogeneous information

Corollary 7.1 (A one-parameter family)**.** *Fix $c > 0$ and $\alpha \geq 0$, and, beginning at a sufficiently large fixed t_0, set*

$$p_t = \frac{c}{t(\ln t)^\alpha} \in (0, 1/2].$$

Then the exact-seed classification and optimal finite-horizon growth are as follows:

Exponent	*Conclusion*
$0 \le \alpha \le 1$	*No countably supported exact infinite seed.*
$1 < \alpha < 2$	*Countable, infinite-entropy seed;* $C_n = \dfrac{c}{(2-\alpha)\ln 2}(\ln n)^{2-\alpha} + O(1)$.
$\alpha = 2$	*Countable, infinite-entropy seed;* $C_n = c \log_2 \ln n + O(1)$.
$\alpha > 2$	*A finite-entropy countable seed exists and* C_n *converges to a finite limit.*

Proof. For all sufficiently large t,

$$h_2(p_t) = \frac{c}{\ln 2}\frac{(\ln t)^{1-\alpha}}{t} + O\left(\frac{1+\ln\ln t}{t(\ln t)^{\alpha}}\right).$$

For $\alpha > 1$ the remainder is summable; the integral test gives $\sum_{t\le n}(\ln t)^{1-\alpha}/t = (\ln n)^{2-\alpha}/(2-\alpha) + O(1)$ when $1 < \alpha < 2$, and $\ln\ln n + O(1)$ when $\alpha = 2$. For $\alpha > 2$ the leading series also converges. Conversely, $\sum p_t$ diverges precisely when $\alpha \le 1$. The statements follow from Theorems 3.1 and 4.1; the existence of the finite limit when $\alpha > 2$ follows from monotonicity and boundedness of C_n. □

8 Why losslessness is not simply the Shannon lower bound

The envelope p_t controls the three phases, but it need not determine C_n exactly. This matters when the word "compression" is used.

Proposition 8.1 (Same envelope, different exact costs)**.** *Fix* $0 < p \le 1/2$. *In the one-step process whose only available success probabilities are* 0 *and* p*, the minimum exact common-seed entropy is* $C_1 = h_2(p)$. *If an additional action with success probability* $p/2$ *is available, then* $C_1 > h_2(p)$. *In particular, at* $p = 1/2$ *the two-action model has* $C_1 = 1$ *bit, while adding the action* $q = p/2 = 1/4$ *raises the exact optimum to* $C_1 = 3/2$ *bits.*

Proof. For the first process use $S \sim \text{Bernoulli}(p)$, returning S for the action with probability p and zero for the other action. The maximizing-action entropy lower bound proves optimality. For the extended process, its finitely many action response bits form one finite random vector F. Any exact seed must have entropy at least $H(F)$, and minimizing $H(F)$ over couplings of the prescribed Bernoulli marginals is a continuous minimum on a compact finite simplex. If that minimum equaled $h_2(p)$, the maximizing response bit B would satisfy $H(F \mid B) = 0$, so every other response bit would be a deterministic function of B. Its probability of equaling one would therefore belong to $\{0, p, 1-p, 1\}$, which excludes $p/2$. Thus the attained minimum is strictly greater than $h_2(p)$. For the explicit $p = 1/2$ example, a joint law for the responses (B, C) with marginals $(1/2, 1/4)$ is parametrized by $x = \mathbb{P}(B = 1, C = 1) \in [0, 1/4]$. Its four cell probabilities are $x, 1/2 - x, 1/4 - x, 1/4 + x$. Shannon entropy is concave in x, so its minimum occurs at an endpoint. Both endpoints give the three positive masses $(1/2, 1/4, 1/4)$ and entropy $3/2$. □

This is also why the upper bound of Theorem 3.1 should not be presented as the exact optimum, nor its constant $2\sum p_t$ as universally sharp. In the critical model it gives the exact leading coefficient, but the bounded correction to C_n remains unspecified.

9 Relation to preceding work and formal-verification boundary

Minimum-entropy couplings of finitely many marginals have an established literature, including Kovačević–Stanojević–Šenk [1] and the profile method of Compton–Katz–Qi–Greenewald–Kocaoglu [2], whose multi-marginal additive constant equals $(1 + \log_2 e)/2$. These results concern marginal couplings; the *shared-history* restriction here specifies a family of causal readouts on one controlled tree. Functional-representation approaches, e.g. Li–El Gamal [3], are another relevant point of

comparison; their setting and optimization target differ from ours. We do not claim that the present elementary sparse-envelope trichotomy, its atomic-capacity formulation, or the last-event identity is a first publication. Priority requires more comprehensive comparison.

The public development [4] currently records formal proofs of the scalar identities $H(G) = 2$, $H(Z_t) = h_2(p_t) + 2p_t$, the finite excess identity, and $h_2(p) \geq p$ on $(0, 1/2]$, in the module `Vanishing.lean` under the `CausalSeed` directory. It also contains a broader formal theory of finite controlled trees, atomic capacity and infinite-horizon seeds. The following results in this note are *mathematical proofs, not new Lean-kernel certificates*: Theorems 3.1, 4.1, 5.1, 5.2, 6.2, and their corollaries. This distinction must be preserved in any repository announcement or academic submission.

Formalization plan. A reliable continuation has four separate interfaces. (1) Model (1) as a history-indexed family of binary kernels, *including zero probabilities*. (2) Define the full strategy readout of independent geometric-gated coordinates and prove cylinder exactness by conditional independence, independently of any entropy calculation. (3) Establish the uniform finite-tail approximation and atomic-capacity bounds; to reuse the current `ITree` and `AtomicCapacity` development, explicitly bridge its positive kernel assumptions to the present zero-branch setting rather than silently applying a theorem beyond its hypotheses. (4) Prove the analytic expansion (12), its summable error, and the last-event identity, then compose the results into one trichotomy.

10 Conclusion

A single common causal seed can reproduce an entire family of adaptive laws without ambiguity at shared histories. Under a sparse attainable envelope, three properties separate sharply:

exact common representation | countable support | finite expected code length.

The first is provided by the gated construction in every regime; the second holds exactly when $\sum p_t < \infty$; the third holds exactly when $\sum h_2(p_t) < \infty$. In the critical case, there is almost-sure global cessation across all policies, yet every exact finite-description model has infinite average information cost; the finite-horizon optimum grows as $\log_2 \ln n + O(1)$. The missing information concerns the times of exceptionally rare events, not an infinite number of realized events.

References

[1] M. Kovačević, I. Stanojević, and V. Šenk, "On the entropy of couplings," *Information and Computation* **242** (2015), 369–382. https://doi.org/10.1016/j.ic.2015.04.003.

[2] S. Compton, D. Katz, B. Qi, K. Greenewald, and M. Kocaoglu, "Minimum-Entropy Coupling Approximation Guarantees Beyond the Majorization Barrier," *Proceedings of AISTATS*, PMLR **206** (2023), 10445–10469. https://proceedings.mlr.press/v206/compton23a.html.

[3] C. T. Li and A. El Gamal, "Strong Functional Representation Lemma and Applications to Coding Theorems," *IEEE Transactions on Information Theory* **64** (2018), 6967–6978. https://arxiv.org/abs/1701.02827.

[4] J. Mikulík, "One Seed for Every Adaptive Policy," Lean 4 development and notes, https://github.com/jendamikulik/finite-Shannon-theorem, consulted 20 October 2026.